Homework Assignment 6: Counting

Due: Sunday, November 15 at 11:59 p.m.

Instructions. Submit your answers as a PDF on Webcourses. I encourage you to work in groups and discuss the problems with one another. You may also use AI tools, but remember that AI will not be available during exams. Use these problems to check your own understanding of the material. Your grade is based on participation, not correctness. You must submit your answers to receive credit for this assignment.

Justify every answer. For counting questions, you may leave answers as exact expressions such as $\binom{n}{k}$ ("n choose k"), factorials, powers, or sums; you do not need to compute their numerical values. Explain why each expression counts the requested outcomes. You may use results proved in lecture, except a result that the question asks you to prove. For Questions 5 and 6, derive the recurrence, give its initial values, and solve it to obtain a closed formula.

Question 1

For each part, say what one outcome is and justify the operation or formula you use.

(a) A shelf has four different novels and three different plays. How many ways can you choose exactly one book? How many ways can you choose one novel *and* one play? Explain why one question uses addition and the other uses multiplication.

(b) There are five shirts, four pairs of pants, and three hats. An outfit has one shirt and one pair of pants, and may have one hat or no hat. How many outfits are possible?

(c) A student counts length-6 binary strings containing at least one 1 as $6 \cdot 2^5$: choose a position for a 1, then fill the other positions freely. Explain the error, give one string counted more than once, and find the correct count.

(d) Count length-8 binary strings that begin with 1 and end with 0. Then count those strings that contain exactly three 1s in total.

(e) Let X have five elements and Y have three elements. How many functions $X \to Y$ are there? How many one-to-one functions $Y \to X$ are there?

(f) Five students are available. Compare the number of ways to appoint a president and a secretary, with distinct people in the roles, with the number of two-person committees. Why is one count twice the other?

Question 2

(a) Eight distinct books are arranged in a row. Count all arrangements. Then count arrangements in which three specified books form one consecutive block, in any order.

(b) How many distinct rearrangements of `BANANA` are there? How many begin and end with `A`?

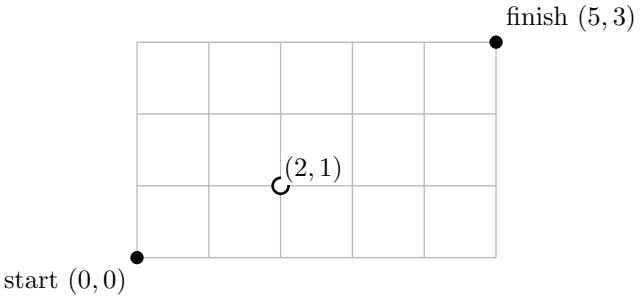
(c) A length-6 password uses $\{A, B, C, D\}$, with repetition allowed. Count those with exactly two As; then count those that also contain at least one B.

Question 3

(a) How many four-person committees can be chosen from eleven people? How many of these committees include a specified person?

(b) A group has eight women and seven men. How many five-person committees contain exactly two women? How many contain at least one woman?

(c) This is a variation of the grid-path question from class. On the five-column, three-row grid below, start at $(0, 0)$, move only right or up along grid lines, and finish at $(5, 3)$. Count all such paths. Then count those that pass through $(2, 1)$ and those that avoid $(2, 1)$.



Question 4

The bins are different, and they may be empty unless a part says otherwise. For distinct balls, which ball goes into each bin matters. For identical balls, only the number of balls in each bin matters.

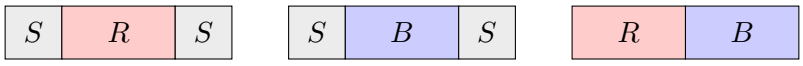
(a) Place seven *distinct* balls into bins A, B, C. Count all ways to put them into the bins, the ways that leave A empty, and the ways that use exactly two bins.

(b) Place nine *identical* balls into bins A, B, C, D. Count all ways to put them into the bins, the ways that give every bin at least one ball, and the ways that give A at least three balls while each other bin gets at least one.

(c) Place ten identical balls into bins A, B, C. How many ways use exactly two nonempty bins? How many give A at most four balls?

Question 5

Let a_n count tilings of a $1 \times n$ board using unit squares of one color and length-2 dominoes available in red and blue. The domino colors are part of the tiling.



Three examples for $n = 4$ (not all the tilings): S is a unit square, R a red domino, and B a blue domino.

(a) Find a_0, a_1. Derive a recurrence for a_n when $n \geq 2$ by examining the leftmost tile. Explain why your cases are disjoint and exhaustive.

(b) Solve the recurrence using its characteristic equation and initial values. Give a closed formula for a_n.

(c) Check your formula at $n = 5$ against the recurrence.

Question 6

Let E_n and O_n count length-n strings over $\{\texttt{A}, \texttt{B}, \texttt{C}\}$ with respectively an even and an odd number of As. The empty string has zero As.

(a) State E_0, O_0. Derive recurrences for both sequences by examining the last symbol.

(b) Eliminate O_n to obtain a recurrence involving only E_n. State the two initial values it needs.

(c) Solve that recurrence using its characteristic equation and give a closed formula for E_n.

Question 7

(a) Find the coefficient of x^4 in $(2x-3)^7$. Show which term of the binomial expansion supplies it.

(b) For $n \geq 0$, use the binomial theorem to find closed forms for

$$\sum_{k=0}^{n} \binom{n}{k} 2^k \qquad \text{and} \qquad \sum_{\substack{0 \leq k \leq n \\ k \text{ even}}} \binom{n}{k} 2^k.$$

(c) For nonnegative integers m, n, r, prove Vandermonde's identity

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k} \binom{n}{r-k},$$

where $\binom{a}{b} = 0$ when $b < 0$ or $b > a$. Give one proof using the binomial theorem and one proof by counting subsets.

Question 8

For each part, define the overlapping sets you count and identify the pairwise and higher intersections that matter.

(a) How many integers from 1 through 200 are divisible by at least one of 2, 3, and 5? Write the full inclusion–exclusion expression for the count.

(b) How many ways can twelve identical balls be put into four labeled bins so that no bin contains more than five balls? Use inclusion–exclusion, starting with all ways to put the balls into bins before the upper bound. Explain why the triple intersections vanish.

(c) Five people each bring a distinct gift, and the gifts are reassigned, one per person. How many assignments give nobody their own gift? Let A_i mean that person i gets their own gift. Explain why an intersection of j specified A_i has $(5-j)!$ assignments, then use inclusion–exclusion.